

Problem A. TLE Khaba

Time Limit 1000 ms

Mem Limit 262144 kB

OS Windows

Statement [Statements \(en\)](#).

The Metropolitan University campus has a long street with n streetlights arranged in a row, numbered from 1 to n . Each streetlight is either on (represented by 1) or off (represented by 0).

The streetlights have a peculiar automatic control system. Every minute, the system updates all streetlights simultaneously according to the following rule:

For a streetlight at position i (where $1 \leq i \leq n$), we define its **neighbors** as all positions j such that $|i - j| = 1$ and $1 \leq j \leq n$.

A streetlight at position i (where $1 \leq i \leq n$) toggles its state (on $\rightarrow$ off or off $\rightarrow$ on) if and only if it has **exactly two neighbors** and **both of them were on** in the previous minute.

You are given the initial state of all streetlights. Your task is to determine the state of all streetlights after exactly k minutes.

Input

The first line contains a single integer T ($1 \leq T \leq 10^5$) — the number of test cases.

Each test case consists of two lines:

- The first line contains two integers n and k ($1 \leq n \leq 10^5, 0 \leq k \leq 10^9$) — the number of streetlights and the number of minutes, respectively.
- The second line contains a binary string s of length n — the initial state of the streetlights, where $s_i = 1$ means the i -th streetlight is on, and $s_i = 0$ means it is off.

It is guaranteed that the sum of n over all test cases does not exceed 10^6 .

Output

For each test case, print a single line containing a binary string of length n — the state of

the streetlights after k minutes.

Examples

Input	Output
1 5 1 01010	01110

Problem B. Hisab koro

Time Limit 1000 ms

Mem Limit 262144 kB

OS Windows

Tariq needs to solve the following math problem for his homework:

"I have 178632754 apples currently. I ate 76987266 apples yesterday. How many apples do I have currently? Provide only the numerical answer."

He decided to consult with *ChatGPT* for help.

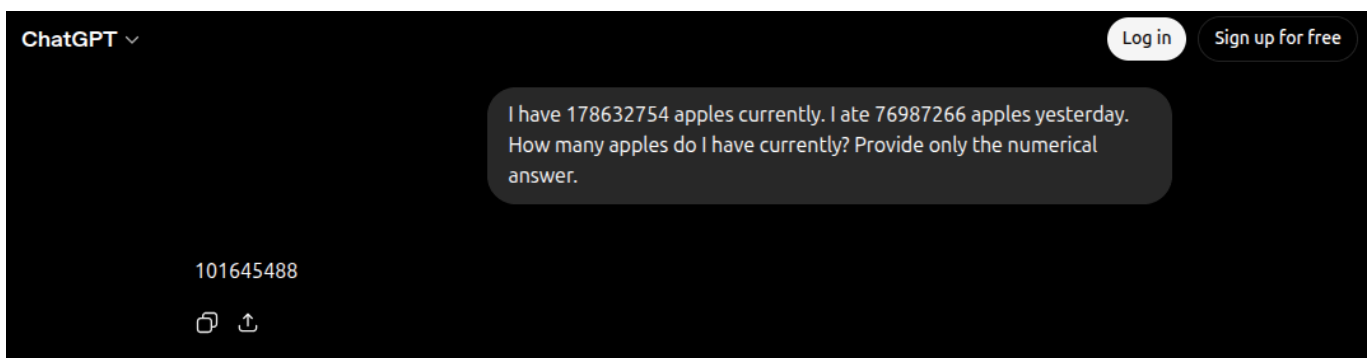


Figure: Conversation with *ChatGPT*. May 20, 2026, 10:07 AM.

However, Tariq does not fully trust *ChatGPT*'s answer, so he came to you. Can you provide the correct answer?

Input

This problem has no input.

Output

Output one single integer — the number of apples.

Problem C. Just one Key

Time Limit 1000 ms

Mem Limit 262144 kB

OS Windows

Statement [Statements \(PDF\)_\(en\)](#)

In the late 1950s, the renowned Bengali mathematician Professor Raj Chandra Bose was working to disprove Euler's long-standing conjecture on Latin Squares at Case Institute of Technology. During his visit, he noticed an exceptional undergraduate student attending his graduate level classes. Impressed by his ability, he invited the student, Donald Knuth, to assist with computational tasks.

As part of this work, Knuth needed to construct a permutation of length n , but only a partial version of the permutation is known.

He is given an array a of length n , where some elements are known and others are missing. Missing elements are represented by -1 .

Your task is to help Knuth replace each occurrence of -1 in a with a value from 1 to n such that:

- The final array a becomes a valid permutation of integers from 1 to n .
- The value of the following expression is minimized:

$$\sum_{i=2}^n (a_i - a_{i-1})$$

If multiple valid permutations achieve the minimum value, print any of them.

Note: A permutation of length n is an array of n distinct integers from 1 to n , each appearing exactly once. For example, $[1]$, $[4, 3, 5, 1, 2]$, and $[3, 2, 1]$ are permutations, while $[1, 1]$ and $[4, 3, 1]$ are not.

Input

The first line contains an integer t ($1 \leq t \leq 10^4$), the number of test cases. The description of the test cases follows.

For each test case, the first line contains an integer n ($2 \leq n \leq 2 \cdot 10^5$), the length of the array.

The second line contains n integers $a_1, a_2, \dots, a_n$ ($-1 \leq a_i \leq n$, $a_i \neq 0$), where $a_i = -1$ denotes a missing value.

It is guaranteed that all positive values in a are distinct.

The sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, print a single line containing n space-separated integers, representing the completed permutation.

If there are multiple valid answers, print any of them.

Examples

Input	Output
2 3 -1 3 -1 5 -1 2 -1 -1 3	2 3 1 5 2 4 1 3

Note

For the first test case, the missing numbers are $\{1, 2\}$.

We try all valid completions:

- $[1, 3, 2] \Rightarrow (3 - 1) + (2 - 3) = 1$
- $[2, 3, 1] \Rightarrow (3 - 2) + (1 - 3) = -1$

The second permutation gives the minimum value, so the best answer is $[2, 3, 1]$.

Problem D. Onk Easy

Time Limit 1000 ms

Mem Limit 262144 kB

OS Windows

Statement [Statements \(PDF\)_\(en\)](#)

You are a close friend of **Dr. G**, a bioinformatics scientist working at **Phitron**. Dr. G is studying the evolution of a synthetic genome in a controlled laboratory environment. Each genome is represented by a non-negative integer, where each bit in its binary representation corresponds to the presence or absence of a specific genetic marker.

The genome evolves over discrete time steps according to the following rules:

- At time step 1, the genome has value a .
- At time step 2, the genome has value b .
- For every time step $t \geq 3$, the genome is formed by examining the genomes at time steps $t - 1$ and $t - 2$, and constructing a new genome that contains only those genetic markers that appear in both of them.

Dr. G has been manually checking whether the genome sequences evolve correctly, but this task has become extremely tedious and time-consuming. To help your friend, you decide to write a program that determines the genome value at a given time step t .

Input

The first line contains an integer T ($1 \leq T \leq 10^5$), the number of independent experiments.

Each of the next T lines contains three integers a , b , and t ($0 \leq a, b \leq 10^6$, $1 \leq t \leq 10^8$), where:

- a is the genome value at time step 1,
- b is the genome value at time step 2,
- t is the time step to be evaluated.

Output

For each experiment, output a single integer — the genome value at time step t .

Examples

Input	Output
3 11 13 1 11 13 2 11 13 3	11 13 9

Note

In this example, the initial genome values are $a = 11$ and $b = 13$.

- For $t = 1$, the genome value is defined as a , so the output is 11.
- For $t = 2$, the genome value is defined as b , so the output is 13.
- For $t = 3$, the genome is constructed by keeping only the genetic markers that appear in both genomes at time steps 1 and 2.

The binary representations of the initial genomes are:

- $11 = (1011)_2$
- $13 = (1101)_2$

Only the leftmost and the rightmost markers are present in both genomes. Therefore, the resulting genome value at time $t = 3$ is: $(1001)_2 = 9$

Problem E. Ektu Besi

Time Limit 1000 ms

Mem Limit 262144 kB

OS Windows

Statement [Statements \(PDF\)_\(en\)](#)

Upobir is playing a game on an infinite 2D grid. He starts at the origin $(0, 0)$ and wants to travel to the target coordinate (x, x) . To navigate the grid, Upobir can perform two types of moves:

- **Chess Knight Move:** From his current position (x_1, y_1) , he can move to any coordinate (x_2, y_2) such that $|x_1 - x_2| = 1$ and $|y_1 - y_2| = 2$, or $|x_1 - x_2| = 2$ and $|y_1 - y_2| = 1$. This move costs 1 unit of energy.
- **Special Jump:** He can perform a unique jump to instantly teleport from his current position directly to the target (x, x) . This unique move does not follow standard chess rules and costs k units of energy.

Given x and k , find the minimum total energy Upobir needs to spend to reach (x, x) .

Input

The first line contains a single integer t ($1 \leq t \leq 10^5$) — the number of test cases.

The only line of each test case contains two integers x and k ($1 \leq x \leq 10^9, 1 \leq k \leq 10^9$) — the coordinate value defining the target (x, x) , and the energy cost of the special jump.

Output

For each test case, print a single integer — the minimum total energy required to reach (x, x) .

Examples

Input	Output
3 1 5 3 2 3 3	2 2 2

Note

In the first test case, Upobir can reach $(1, 1)$ using 2 chess knight moves: $(0, 0) \rightarrow (-1, 2) \rightarrow (1, 1)$. This costs 2 units of energy.

In the second test case, Upobir can reach $(3, 3)$ using 2 chess knight moves: $(0, 0) \rightarrow (1, 2) \rightarrow (3, 3)$. Alternatively, he could use the special jump immediately at $(0, 0)$ to instantly leap to $(3, 3)$. In both scenarios, the minimum energy required is 2.

Problem F. Corner Case

Time Limit 1000 ms

Mem Limit 262144 kB

OS Windows

Statement [Statements \(PDF\)_\(en\)](#)

You are given a string s consisting of lowercase English letters.

Find two non-empty substrings* p and q of s such that p is not a substring of q , and q is not a substring of p . Note that p and q do not have to be disjoint — they may correspond to overlapping positions in s . Over all such pairs (p, q) , output the maximum possible value of $|p| + |q|$, where $|x|$ denotes the length of a string x .

If no such pair exists, output 0.

*A string a is a substring of a string b if a can be obtained from b by deletion of several (possibly, zero or all) characters from the beginning and several (possibly, zero or all) characters from the end. For example, `us` is a substring of `sust`, but `ut` and `aus` are not.

Input

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

The first and only line of each test case contains a string s ($1 \leq |s| \leq 3 \cdot 10^5$) consisting of lowercase English letters.

It is guaranteed that the sum of $|s|$ over all test cases does not exceed $3 \cdot 10^5$.

Output

For each test case, output a single integer — the maximum value of $|p| + |q|$ over all valid pairs, or 0 if no valid pair exists.

Examples

Input	Output
2 ab z	2 0

Note

In the first test case, $s = \text{ab}$. Choose $p = \text{a}$ and $q = \text{b}$; neither string is a substring of the other, and $|p| + |q| = 1 + 1 = 2$, which is the maximum possible.

In the second test case, $s = \text{z}$ has only one substring, z itself, so the only possible choice is $p = q = \text{z}$. But then p is a substring of q , so this pair is not valid. No valid pair exists, and the answer is 0.